

Edge-Based Defense in Networks

A Modification of Bloch, Chatterjee & Dutta (2023)

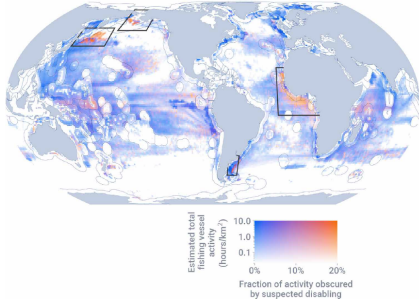
Chukwudi Samuel Ogbuta

Supervised by

Prof. Sylvain Béal • Prof. Emmanuel Peterlé

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Strategic adversaries on networks



Estimated global fishing activity. Orange marks suspected illegal activity — West Africa is one of three concern zones.

Welch et al., *Science Advances* (2022) • Global Fishing Watch

The problem.

Illegal fishing reaches places no single country owns. Vessels move along sea routes between national waters and high seas, taking what isn't theirs to take.

The adversary is strategic.

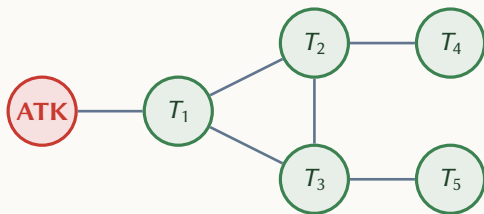
Routes are chosen, not random. The weakest gap is the one used.

The defenders are many.

Neighbouring states watch the same waters. Defense is naturally shared.

What is a network?

A network is a graph $G = (V, E)$: nodes V are locations, edges E are direct connections between them.



Nodes.

Locations the attacker can reach.

Edges.

The connections between locations.

The attacker.

A single adversary outside the network.

The defenders.

One at each target. They invest to intercept.

Two traditions, one gap

	Node Defense		Edge Defense
	Centralized	Continuous	Centralized
<i>Authors</i>	Dziubiński & Goyal Cerdeiro et al.	Bloch, Chatterjee & Dutta (2023)	Washburn & Wood Mavronicolas et al.
<i>Investment</i>	Binary	Continuous	Discrete or budgeted
<i>Defenders</i>	Single planner	Many	Single planner

The gap — *Continuous, decentralized investment — placed on the connections.*

The Bloch baseline: setup

$$\alpha_{k(p)}(p) = \prod_{\substack{j \in V(p) \\ j \neq 0, j \neq k(p)}} (1 - x_j)$$

$$U(\pi, x) = \sum_{p \in P} \pi(p) \beta_{k(p)}(p) b_{k(p)}$$

$$V_i(\pi, x) = - \sum_{\substack{p \in P \\ k(p)=i}} \pi(p) \beta_i(p) b_i - \frac{x_i^2}{2}$$

Path survival.

The attacker survives the interior of the path if it slips past every defender between the attacker node and the target.

Attacker's expected payoff.

Sum over every path, weighted by how often the attacker uses it, times the chance of getting through, times the target's value.

Defender's payoff.

Expected loss from attacks on its own target, minus the cost of its investment. Defenders are the only players who pay a cost.

The Bloch baseline: equilibrium

Bloch distinguishes two kinds of target. A first target is reached directly from the attacker node; a deeper target is reached only by passing through other targets that are themselves attacked.

$$x_i = 1 - \frac{U}{b_i}, \quad q_i = \frac{1}{b_i} \left(1 - \frac{U}{b_i} \right)$$

$$x_i = 1 - \frac{b_k}{b_i}, \quad q_i = \frac{b_k}{b_i U} \left(1 - \frac{b_k}{b_i} \right)$$

First-target equilibrium.

Higher-value targets invest more and are attacked more often. U is the common payoff across all attacked targets.

Deeper-target equilibrium.

For a target i reached only through its predecessor k . The predecessor's value b_k replaces U as the relevant threshold.

What Bloch leaves on the table

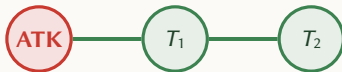
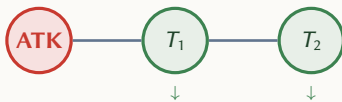
Defense lives at the location.

In Bloch, every defender's investment sits at its own node. The attacker is caught *at* a target, not on the way to one.

But corridors are shared.

A shipping lane, a forest road, a fishing zone — none of these belong to one party alone. Real defense happens *on the way*, by whoever watches that stretch.

Bloch: defense at nodes



Edge-based: defense on edges

The modification, stated

Move investment from nodes to edges. Every edge has two defenders.

Bloch's baseline

$$S(\text{node } i) = 1 - x_i$$

One defender per node. Survival is whatever it fails to catch.

The modification

$$S(e) = (1 - x_{a,e})(1 - x_{b,e})$$

Two defenders per edge. The attacker escapes only if *both* sides fail.

Cost is split: each defender i pays $\sum_{e \ni i} x_{i,e}^2 / 2$, summed over its incident edges.

What carries over from Bloch

Three of the four key lemmas in Bloch's analysis depend only on the attacker's optimization. They survive the move to edges.

Equal payoff across the support

Every attacked target gives the attacker the same payoff U .

Optimal predecessor has lowest value

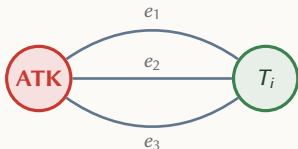
If a target can be reached through several support members, the attacker passes through the one with the lowest value.

Values increase along attack paths

Target values rise as the attacker moves deeper. Survival decay is offset by rising value.

What changes (1): the multi-path factor

A target reachable through several free-highway edges is structurally weaker: each edge must be defended separately.



*m_i = 3 free-highway edges
reaching the same target*

$$\begin{aligned} \text{Bloch: } q_i &= \frac{b_i - U}{b_i^2} \\ \text{Edge-based: } q_i &= \frac{m_i (b_i - U)}{b_i^2} \end{aligned}$$

The factor m_i counts the free-highway edges feeding the target.
More edges $\Rightarrow$ more attack probability $\Rightarrow$ weaker target.

What changes (2): strategic substitutes

When two defenders share an edge, one's investment changes the other's incentive.

$$x_{i,e}^* = q b_i (1 - x_{j,e})$$

$$\frac{\partial x_{i,e}^*}{\partial x_{j,e}} = -q b_i < 0$$

If my neighbor invests more, I invest less.

The slope is strictly negative whenever attacks flow through the edge. Defenders' efforts substitute for each other.

But the edge cannot stay undefended.

If the other side invests zero, my best response is strictly positive. $(0, 0)$ is never an equilibrium when attacks reach the edge.